

Selection (σ) Operator in Relational Algebra

EXAM-READY STUDY NOTES & REVISION GUIDE

1 What is Selection?

Selection is a fundamental unary operator in Relational Algebra represented by the Greek letter σ (**Sigma**).

★ EXAM-READY DEFINITION

Selection (σ) is a unary relational algebra operation that selects tuples (rows) from a relation that satisfy a given selection condition or predicate.

Syntax:

$$\sigma_{condition}(Relation)$$

Example:

$$\sigma_{Roll_No = 2}(Student)$$

Meaning: Select all rows from the **Student** relation where **Roll_No** = 2.

2 Selection Example

Consider the **Student** table:

Roll_No	Name	Age
1	A	20
2	B	21
3	A	20

Query: $\sigma_{Roll_No = 2}(Student)$

Roll_No	Name	Age
2	B	21

3 Selection Works on Rows ★★★★★

This is the most fundamental property to remember: **Selection filters horizontally across rows (tuples)** based on the specified condition.

Consider the Student table with updated data:

Roll_No	Name	Age
1	A	20
2	B	21
3	C	22

Query: $\sigma_{Age > 20} (Student)$

Roll_No	Name	Age
2	B	21
3	C	22

4 Combining Selection & Projection

Many real-world exam questions require combining row filtering (Selection) and column extraction (Projection).

Question: Retrieve the name of the student whose Roll_No = 2.

- **Condition:** Roll_No = 2 (Requires Selection σ)
- **Final Output Column:** Name (Requires Projection π)

Step 1: Apply Selection (σ)

$$\sigma_{Roll_No = 2} (Student)$$

Intermediate Result: Row for Roll_No = 2 (all columns retained).

Step 2: Apply Projection (π)

$$\pi_{Name} (\sigma_{Roll_No = 2} (Student))$$

Name
B

5 Order of Operations (Why Selection Comes First)

If you perform projection first: $\pi_{Name} (Student)$, the output only contains the Name column. The Roll_No column is discarded!

Attempting to run $\sigma_{Roll_No = 2}$ afterwards fails because Roll_No no longer exists in the intermediate relation.

 **Golden Rule: First Select Required Rows (σ), Then Project Required Columns (π).**

6 Selection (σ) vs. Projection (π)

Feature	Selection (σ)	Projection (π)
Symbol	σ (Sigma)	π (Pi)
Works On	Rows (Tuples)	Columns (Attributes)
Term	Tuple Selection	Attribute Selection
Based On	Predicate / Condition	Attribute List
Example	$\sigma_{Age > 20} (Student)$	$\pi_{Name} (Student)$
Duplicate Removal	No duplicate elimination	Eliminates duplicate tuples

 **Memory Trick:** σ = Rows (Select) | π = Columns (Pick)

7 Additional Example (Multiple Output Attributes)

Question: Retrieve the Name and Age of students whose Roll_No = 2.

$\pi_{Name, Age} (\sigma_{Roll_No = 2} (Student))$

Name	Age
B	21

8 Properties of Selection

- **Non-Destructive:** Selection produces a new result relation. The original table in the database is not modified or deleted.
- **Unary Operator:** Operates on a single input relation at a time.

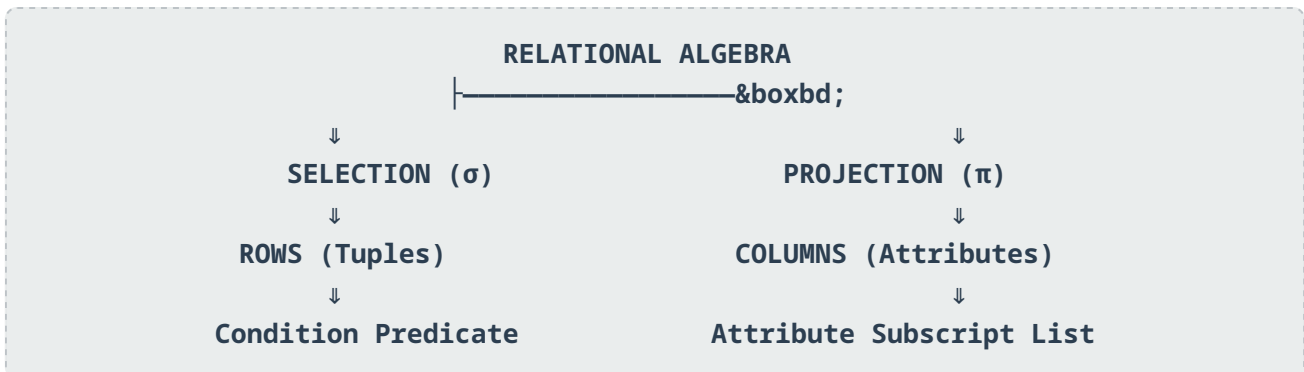
COMMON EXAM TRAPS

- **Trap 1:** Thinking Selection operates on columns (It operates on *rows*).
- **Trap 2:** Thinking Projection operates on rows (It operates on *columns*).
- **Trap 3:** Executing projection before selection when the selection attribute is projected out (Always *select first*).

✓ Key Exam Takeaways & Summary

- **Symbol:** Selection is denoted by σ (Sigma).
- **Target:** Filters **rows** / **tuples** based on a boolean condition.
- **Operator Type:** Unary (takes one relation as input).
- **Duplicates:** Does not eliminate duplicate tuples automatically.
- **Query Order:** Filter rows using σ before projecting columns using π .

🚀 Quick Revision Diagram



One-Line Summary: Selection (σ) is a unary relational algebra operation that selects tuples/rows from a relation satisfying a specified condition.